



Yamin Haque

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Mirpur, Dhaka 1216

PROFESSIONAL SUMMARY

Embedded Software Engineer with 3+ years of experience developing, debugging, and maintaining C/Embedded C software for ARM-based systems running Embedded Linux and VxWorks. Experienced in firmware development, MCU architecture, hardware/software interfacing, peripheral communication, and low-level debugging. Hands-on experience with STM32 microcontrollers, GPIO, ADC, SPI, I2C, UART, EXTI, timers, and sensor integration through bare-metal firmware development. Strong background in embedded networking, protocol implementation, system-level troubleshooting, and standards-based development. Experienced with GDB, GCC, Wireshark, kernel tracing, and command-line development workflows. Comfortable collaborating with cross-functional and overseas engineering teams to develop, test, debug, and deliver reliable embedded products.

CORE EMBEDDED SKILLS

- **Firmware:** Embedded C, C++, Bare-metal Programming, Register-level Programming, Firmware Debugging
- **MCU / Architecture:** STM32F4, ARM Cortex-M, MCU Architecture, Processor Low-power Modes
- **Peripherals:** GPIO, ADC, SPI, I2C, UART, EXTI, Timers, Ethernet
- **Embedded OS:** Embedded Linux, VxWorks, FreeRTOS, Linux
- **Debugging / Testing:** GDB, GCC, Wireshark, trace-cmd, KernelShark, ftrace, Scapy
- **Communication:** UART, SPI, I2C, TCP/UDP, MQTT, HTTP/HTTPS
- **Version Control:** Git, SVN

WORK EXPERIENCE

Software Engineer (R&D)

Apr 2023 – Jul 2026

Shanghai BDCOM Information Technology Co., Ltd. (Bangladesh Office)

Rangs Paramount-2, Level-2, Block-K, Plot-11, Road-17, Banani, Dhaka-1213

- Developed, implemented, tested, and maintained embedded C software for ARM-based Carrier Ethernet systems running Embedded Linux and VxWorks.
- Designed and maintained Layer 2 networking firmware including LLDP, ERPS (G.8032), EAPS, CFM/OAM, VLAN, MSTP, and SNMP using GCC, GDB, and command-line development workflows.
- Studied and implemented IEEE, ITU-T standards, and relevant RFCs to achieve protocol compliance, interoperability, and reliable system operation.
- Performed low-level debugging and root-cause analysis involving pointers, memory management, packed structures, byte-order conversion, 64-bit counters, and protocol behavior using GDB, Wireshark, trace-cmd, KernelShark, and ftrace.
- Analyzed hardware/software interactions and event-driven system behavior while troubleshooting embedded networking and system-level issues.
- Collaborated with cross-functional teams and overseas R&D engineers to troubleshoot field issues, improve system stability and convergence, resolve interoperability problems, and deliver customer-specific embedded features.

EMBEDDED PROJECT EXPERIENCE

Bare-Metal STM32F446RE Thermostat Controller ([GitHub](#))

- Developed bare-metal embedded firmware in C for the **STM32F446RE ARM Cortex-M4 MCU** using direct register-level programming.
- Configured and interfaced **GPIO, ADC, I2C, SPI, UART, EXTI, and timer peripherals** without relying on high-level peripheral libraries.
- Integrated a **BME280 environmental sensor over I2C** and **SSD1306 OLED over SPI** for real-time temperature/environment monitoring and display.

- Implemented RUNNING/CONFIG operating modes, push-button interrupt handling, potentiometer-based temperature setpoint control, and GPIO-based heating indication.
- Validated peripheral initialization, sensor communication, interrupt handling, and firmware control logic through hardware-level testing and debugging.

TFTP Server Module([GitHub](#))

- Developed a standalone TFTP server implementing **RFC 1350** for file transfer between network devices and management systems.
- Implemented configurable UDP port, timeout, retransmission retry count, and server enable/disable controls through a CLI.
- Added support for **RFC 2348 Blocksize Option** and **RFC 7440 Window size Option** to improve transfer performance and protocol compliance.

ACADEMIC PROJECT

16-bit CPU & Digital System Design

- Designed and implemented a 16-bit CPU using HDL, including the ALU, register file, program counter, control unit, flag register, and main memory.
- Developed a custom Instruction Set Architecture (ISA) and assembler supporting arithmetic/logic operations, branching, stack, and procedure operations.
- Designed and integrated adder/subtractor, unsigned multiplier, and divider units into the CPU datapath.
- Implemented memory interfacing and validated CPU operation through assembly-level programs.

ADDITIONAL TECHNICAL SKILLS

- **Networking:** Ethernet, TCP, UDP, TFTP, VLAN, STP/RSTP/MSTP, LLDP, ERPS (G.8032), EAPS, SNMP, CFM/OAM
- **Messaging:** MQTT, HTTP/HTTPS
- **Programming:** Embedded C, C++, PHP
- **Development Tools:** Git, SVN, GCC, GDB

EDUCATION

Bachelor of Science in Computer Science & Engineering
Rajshahi University of Engineering & Technology, Bangladesh

2017 – 2022
CGPA: 3.11/4.00

ONLINE ACCOUNTS

- Codeforces: [Yamin1999](#) – Solved 100+ problems
- LeetCode: [Yamin1999](#) – Solved 70+ problems

REFERENCE

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